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1. Introduction

Inter-Process Communication (IPC) is a mechanism that allows processes to exchange data and synchronize their actions.

One common IPC mechanism in Unix/Linux systems is the **Message Queue**.

A message queue allows processes to send and receive messages in a structured and safe manner without sharing memory directly.

This activity demonstrates communication between two processes using a **System V Message Queue**.

2. Objective

The objectives of this activity are:

- To understand Inter-Process Communication (IPC)
 - To use Message Queue for data communication
 - To implement sender and receiver processes
 - To observe message passing between processes
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3. IPC Mechanism Used

Message Queue

A message queue allows messages to be sent and received asynchronously.

Each message has:

- A message type
- Message data

Processes do not need to run at the same time to communicate.

4. Files Description

The activity consists of the following files:

- **common.h** – Contains shared definitions and message structure
- **sender.c** – Sends a message to the message queue
- **receiver.c** – Receives a message from the message queue

5. Program Description

5.1 Sender Program (`sender.c`)

The sender program performs the following steps:

1. Creates or accesses a message queue
2. Prepares a message
3. Sends the message to the queue
4. Terminates after sending

5.2 Receiver Program (`receiver.c`)

The receiver program performs the following steps:

1. Connects to the same message queue
2. Waits for a message
3. Receives and displays the message
4. Terminates after receiving

6. Compilation and Execution

6.1 Compile the Programs

```
gcc sender.c -o sender
gcc receiver.c -o receiver
```